Windows and the DTFT: Trading Resolution for Dynamic Range

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Key Concepts

1) The relatively high side lobes of the rectangular window are a consequence of the abrupt transition at the boundaries of the window.

2) Smoother transitions provided by other types of windows or tapers result in reduced side lobe height at the expense of wider main lobe width.

    a) The main lobe width of the rectangular window is $\frac{4\pi}{N}$ rads while the peak side lobes are -13 dB relative to the main lobe. The rectangular window has the narrowest possible main lobe.

b) The main lobe width of the Hamming window is $\frac{8\pi}{N}$ rads while the peak side lobes are –41 dB relative to the main lobe.

c) The Chebyshev window has equal side lobes with height specified by the user and the smallest possible main lobe width for the desired side lobe height.

d) There are many different windows available that provide different main lobe and side lobe characteristics.

3) Different windows provide a way to trade dynamic range (side lobe height) for resolution (main lobe width).

Video Screenshots

Windows: Trading Resolution for Dynamic Range

$$z[n] = x[n]\,w[n]$$

truncates $x[n]$ to interval $0 \le n \le N-1$

$$Z(e^{j\omega}) = \frac{1}{2\pi} X(e^{j\omega}) * W(e^{j\omega})$$

- High sidelobes in $W(e^{j\omega})$ from sharp transition in $w[n]$

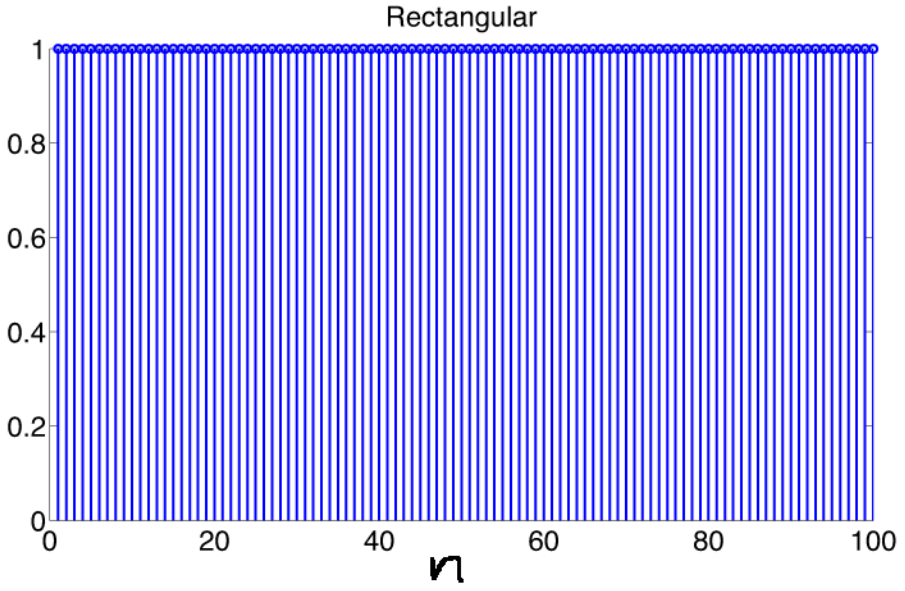
Use different shaped windows (or tapers) to reduce sidelobe height in return for wider mainlobe

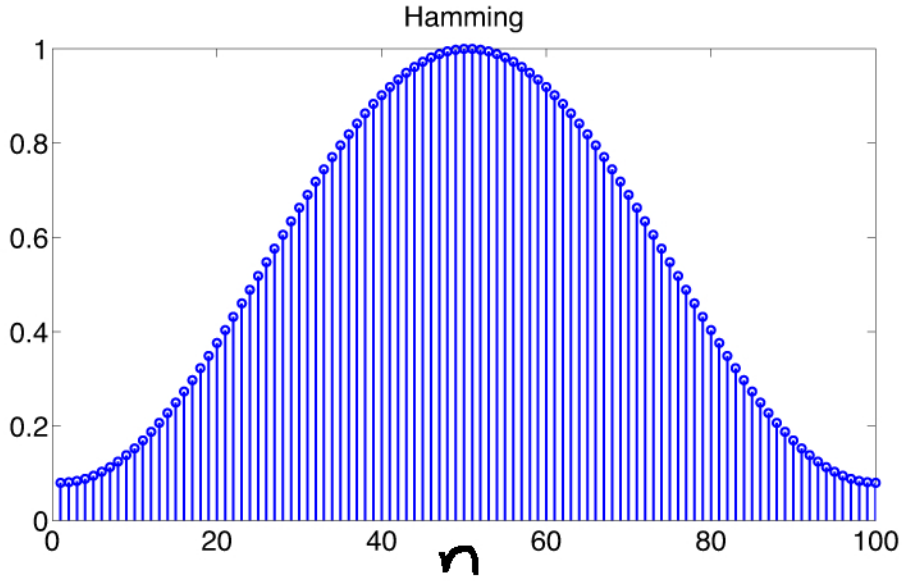
Example: Hamming window

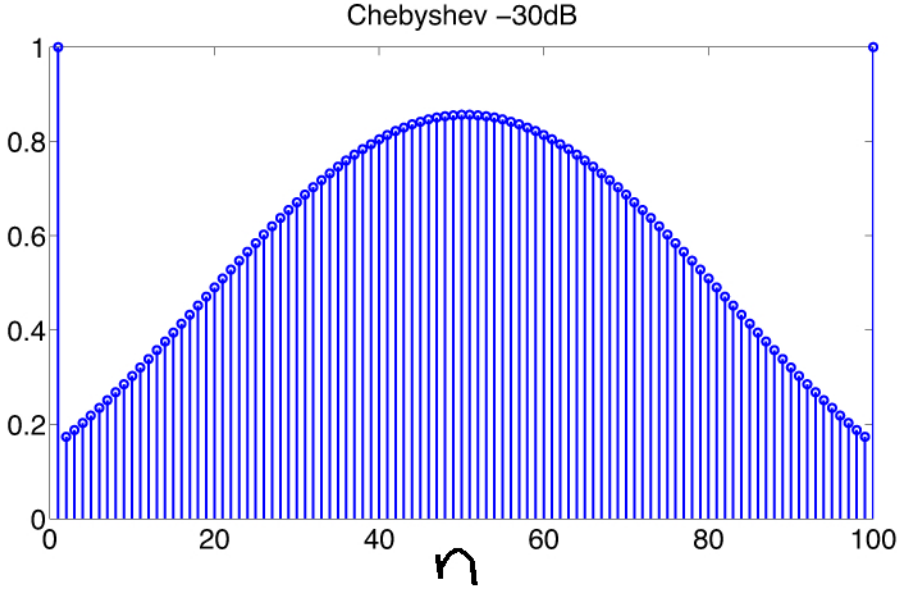
$$w[n] = \begin{cases} 0.54 - 0.46\cos\left(\frac{2\pi n}{N-1}\right), & 0 \le n \le N-1 \\ 0, & \text{otherwise} \end{cases}$$

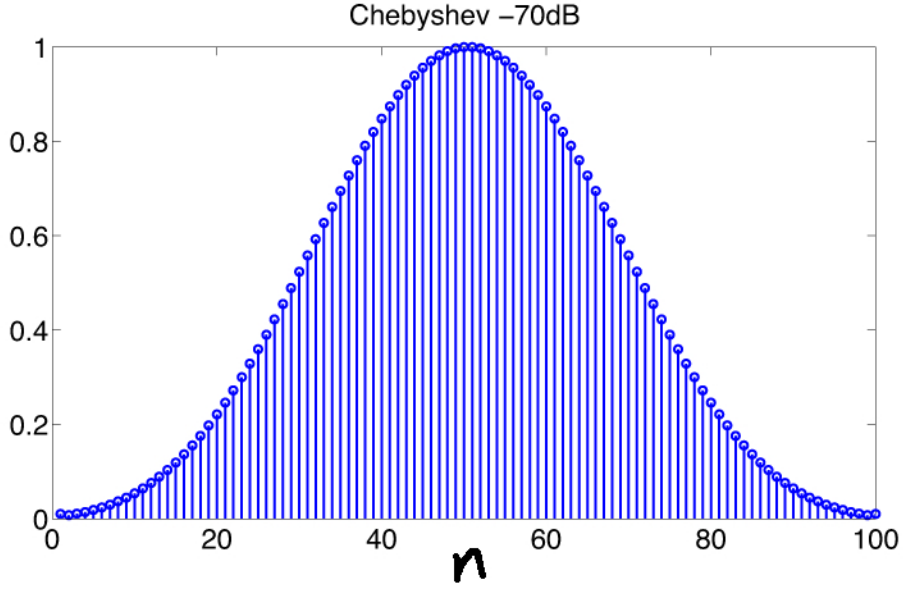


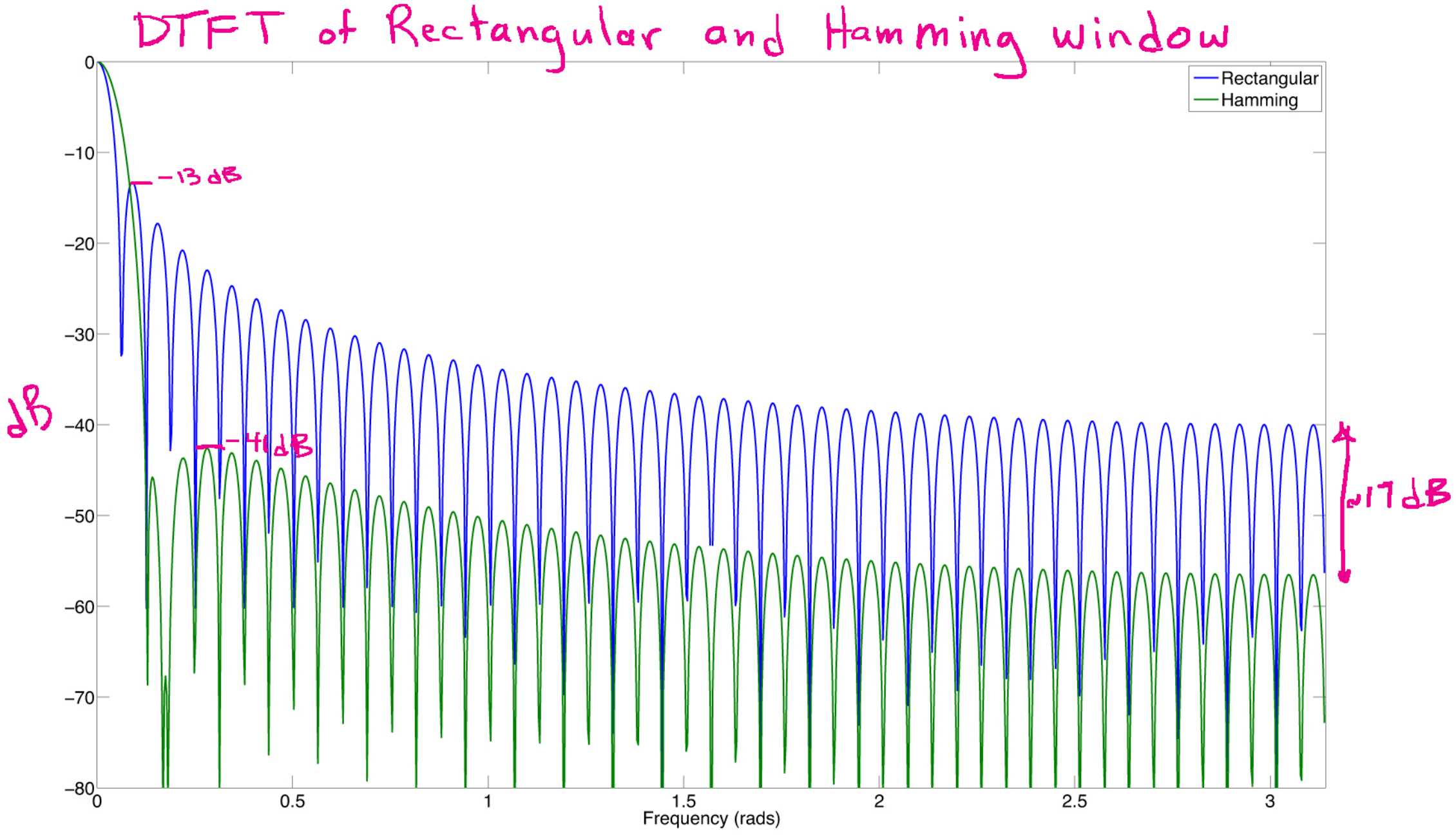
window	mainlobe width	sidelobe height
rectangular	$4\pi/N$	-13 dB
Hamming	$8\pi/N$	-41 dB

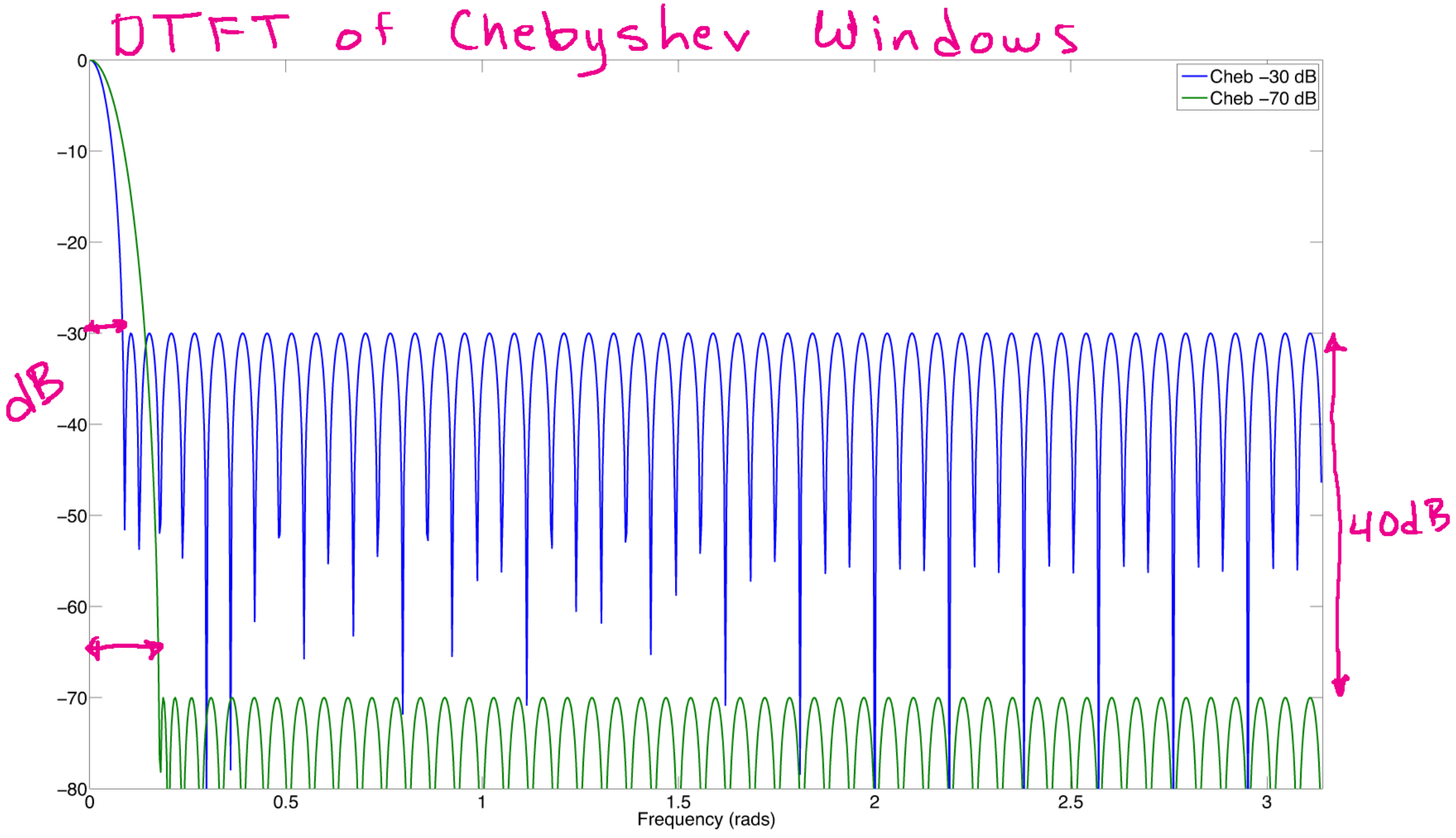

Rectangular
1
0.8
0.6
0.4
0.2
0
0
20
40
60
80
100
n


Hamming
1
0.8
0.6
0.4
0.2
0
0
20
40
60
80
100
n


Chebyshev −30dB
1
0.8
0.6
0.4
0.2
0
0
20
40
60
80
100
n


Chebyshev −70dB
1
0.8
0.6
0.4
0.2
0
0
20
40
60
80
100
n


DTFT of Rectangular and Hamming window
Rectangular
Hamming
dB
0
−10
−20
−30
−40
−50
−60
−70
−80
−13 dB
−41 dB
≈17 dB
0
0.5
1
1.5
2
2.5
3
Frequency (rads)


DTFT of Chebyshev Windows
Cheb –30 dB
Cheb –70 dB
0
–10
–20
–30
–40
–50
–60
–70
–80
dB
40dB
0
0.5
1
1.5
2
2.5
3
Frequency (rads)

Example:

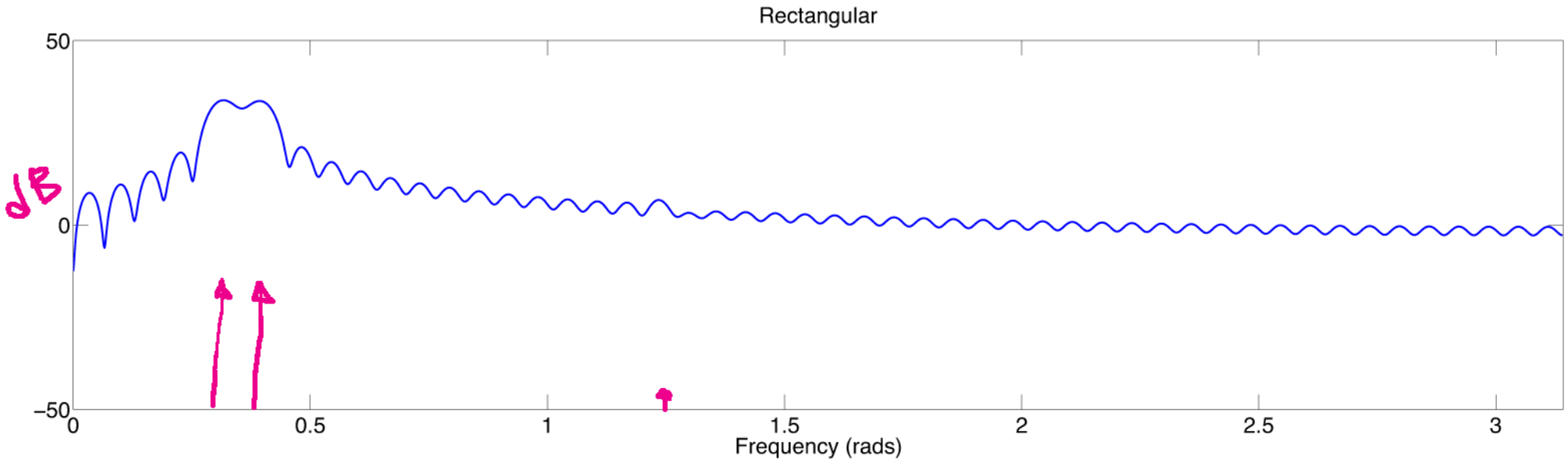
$$x[n] = \cos\left(\frac{\pi}{10}n\right) + \cos\left(\frac{\pi}{8}n + \frac{\pi}{3}\right) + 0.01\cos\left(\frac{4\pi}{10}n - \frac{3\pi}{5}\right)$$

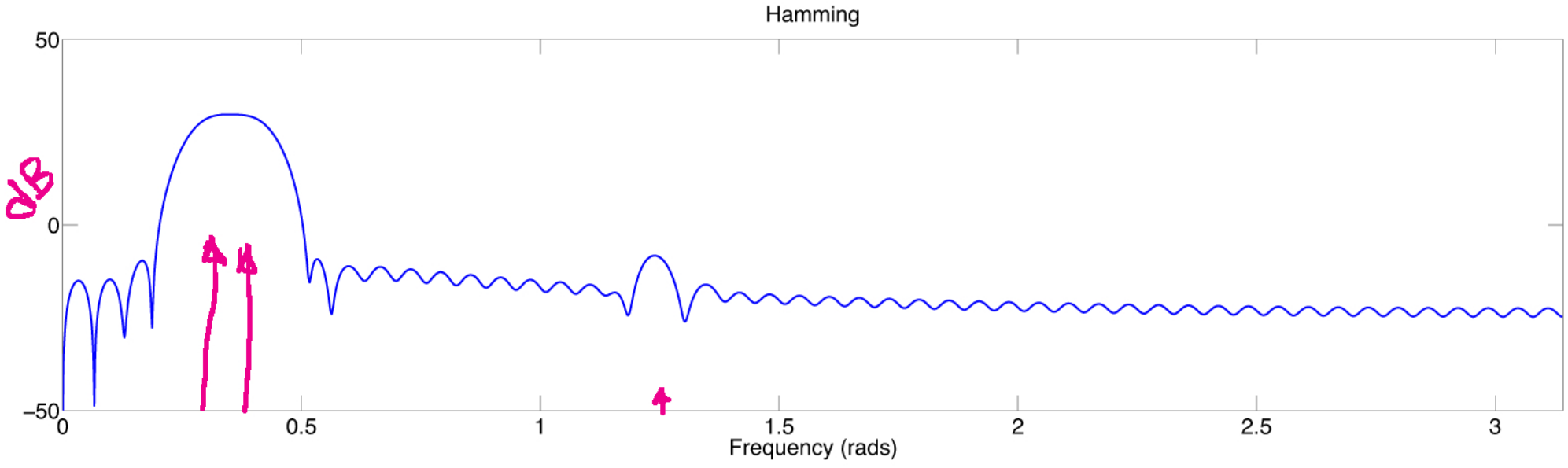
equal amplitude

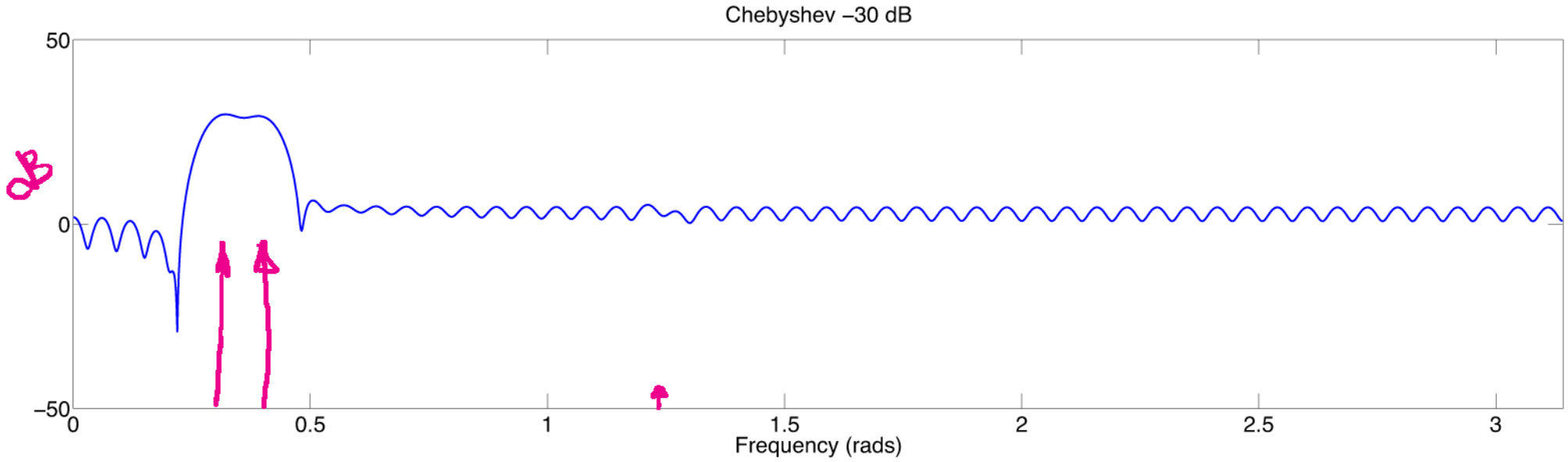
weak amplitude

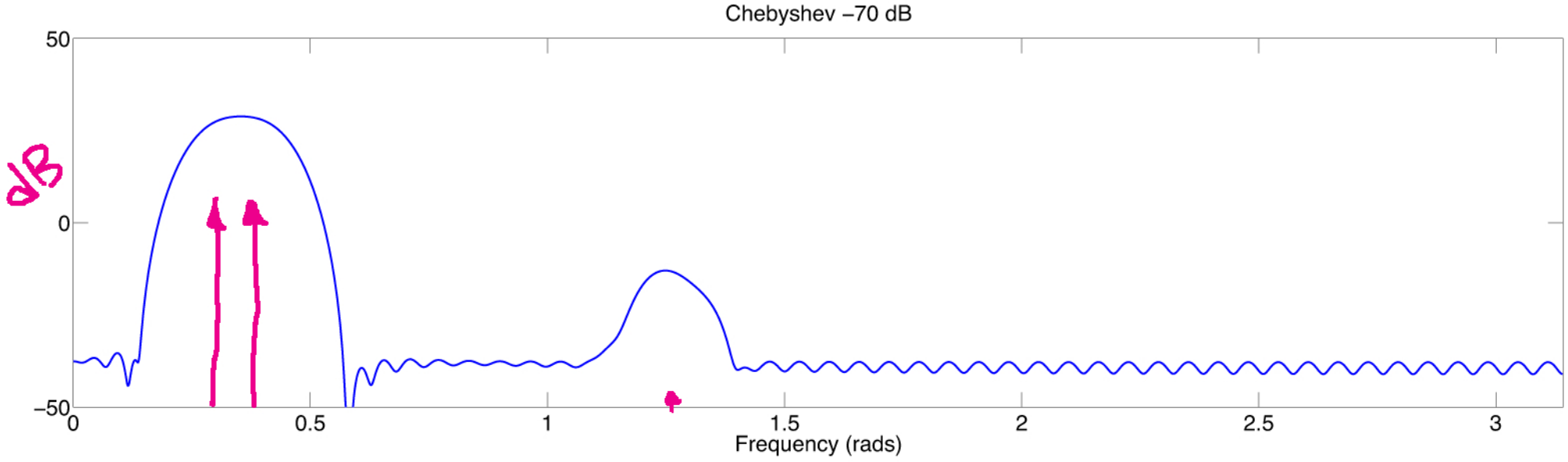
$N = 100$:

1) Rectangular window
2) Hamming window
3) Chebyshev -30 dB
4) Chebyshev -70 dB


Rectangular
50
0
−50
dB
0
0.5
1
1.5
2
2.5
3
Frequency (rads)


Hamming
50
0
−50
dB
0
0.5
1
1.5
2
2.5
3
Frequency (rads)


Chebyshev –30 dB
50
0
–50
dB
0
0.5
1
1.5
2
2.5
3
Frequency (rads)


Chebyshev –70 dB
50
0
–50
dB
0
0.5
1
1.5
2
2.5
3
Frequency (rads)

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